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Black Mamas Matter: Maternal Leadership and COVID-19 Vaccine Decisions Among Black Women in Michigan

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Abstract
Movements such as MAHA (Make America Healthy Again) and the resurgence of anti-vaccine rhetoric promoted by figures like Robert F. Kennedy, Jr. have amplified skepticism, particularly in Black communities already burdened by structural inequities and historical medical racism. These contemporary patterns of resistance reflect a broader continuity between COVID-19 vaccine hesitancy and ongoing distrust toward public health institutions, underscoring the need for culturally grounded engagement strategies that honor lived experience and community leadership. This qualitative study examines how Black women in Michigan made decisions about the COVID-19 vaccine for themselves and their families, and how their choices influenced broader community responses to the vaccine. Using a community-based participatory research (CBPR) approach, we conducted 34 interviews between 2022 and 2023. In the current study we focus on the 20 Black women who were interviewed as part of the larger study. Categorized by Self-Determination Theory constructs, we found that relatedness, through a desire to protect others, was the dominant motivator, expressed through caregiving rooted in cultural traditions. Using the frameworks of Inventive Mothering and Black Mothering, we explore how maternal leadership shaped vaccine acceptance and alternative protective strategies.

Keywords COVID-19 vaccine · Maternal leadership · Self-Determination theory · Inventive mothering · Black mothering · Health equity · Vaccine hesitancy · Michigan

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Introduction

Although this manuscript was conceptualized at the height of the COVID-19 pandemic and is being published after the pandemic has been officially declared over, recent shifts in the political landscape have heightened the urgency of addressing vaccine hesitancy. The increasing visibility of figures like Robert F. Kennedy Jr. and organized movements such as Make America Healthy Again (MAHA) have fueled skepticism not only toward COVID-19 vaccines but also toward long-established childhood immunizations, including the measles-mumps-rubella (MMR) vaccine, hepatitis A and B vaccines, and the seasonal flu vaccine [1–5]. These narratives, often framed as protecting parental rights or bodily autonomy, intersect with long-standing histories of medical mistrust in Black communities, creating layered challenges for public health [6, 7]. Under the current administration, where debates over vaccine mandates, parental choice, and misinformation remain prominent, understanding these expanding forms of hesitancy is crucial. Public health education and outreach strategies must adapt to this new landscape, recognizing that the erosion of trust surrounding one vaccine can spill over to others, further widening disparities in preventive care [8, 9].

During the COVID-19 pandemic, Black Americans experienced higher rates of infection, hospitalization, and death compared to White Americans nationwide [10, 11]. National surveys also showed that Black adults expressed higher levels of vaccine hesitancy compared to White adults, often citing safety concerns, mistrust of government, and a historic legacy of medical racism [12–15]. In Michigan, Black residents represented only 14% of the population but accounted for over 40% of COVID-19 deaths in the early months of the pandemic [16]). Throughout the pandemic, Black Michiganders continued to die from COVID-19 at nearly twice the rate of White Michiganders [17]. In Wayne County, which includes Detroit, the crude COVID-19 death rate reached nearly 179 per 100,000 residents, one of the highest in the state [18]. Within the Detroit City Health District, the age-adjusted mortality rate was 64.4 per 100,000 in 2022, reflecting the enduring toll of the pandemic [19]. Early reports further indicated that Black residents accounted for more than 75% of COVID-19 diagnoses and nearly 90% of deaths in Detroit [20]. These inequities reflect long-standing structural determinants of health, including unequal access to healthcare, higher prevalence of comorbidities, and systemic racism in medical systems [21].

These disparities are not isolated to the pandemic but are embedded within broader historical and structural inequities that have long shaped health outcomes in Black communities. Scholars have documented how structural racism in housing, employment, and healthcare access contributes to disproportionate burdens of chronic disease, reduced life expectancy, and increased vulnerability to infectious disease [22, 23]. Black Americans consistently appear at the top of nearly every major health disparity index, including hypertension, diabetes, maternal mortality, and cancer outcomes [24, 25]. COVID-19 magnified these entrenched inequities by exposing the fault lines of a health system that has consistently failed Black populations. Studies show that during the pandemic, Black adults were less likely to have access to testing [26], expressed higher levels of vaccine hesitancy than White counterparts [13, 14], and often faced treatment delays or reduced quality of care once diagnosed with COVID-19 [27]. These trends suggest that the challenges faced by Black communities during the pandemic are part of an ongoing struggle toward achieving health equity. While much of the public health literature on Black communities emphasizes church-based health promotion, health messaging, and access interventions, a substantial body of evidence also points to the political strength and cultural authority of Black women as a driving force within Black communities [24, 28]. This raises an important question: if Black women's leadership and influence have historically shaped civic and political outcomes, might these same dynamics extend to matters of health equity?

For Black women, the intersection of race, gender, and class situates them as both caregivers and cultural authorities within their families and communities. Prior scholarship highlights that Black women often act as health advocates and decision-makers, not only for themselves but also for partners, children, and extended kin [29]. For example, Allen and colleagues (2018) found that Black women played a central role in guiding their partners' prostate cancer screening decisions [30], while Chatters, Taylor, and Lincoln (1999) documented their disproportionate leadership in church-based social and health activities [31]. Hill further argued that Black women's caregiving and advocacy roles make them indispensable to the overall health of Black communities [32]. These findings align with a broader body of literature indicating that men often rely on women in their families, i.e., mothers, wives, and grandmothers, to encourage preventive health behaviors [33]. Collectively, this research demonstrates that Black women exercise cultural authority over health practices through their roles as mothers, caregivers, and community leaders. Building on this foundation, our study positions Black mothers' COVID-19 vaccine decisions not only as personal choices but as influential acts with implications for household and community health.

Previous research on Black mothers' perspectives regarding COVID-19 vaccination remains limited. Allen and colleagues (2021) found that non-Latino Black women were less likely than non-Latino White women

to report willingness to be vaccinated, underscoring the central role of women as "shepherds" of medical care in many families [34]. More recent qualitative studies in Chicago, Kentucky, and elsewhere have further examined vaccine decision-making among Black women [35, 36]. Harris and colleagues reported that racism in healthcare contributed directly to hesitancy among Black mothers [36]. Mohammed and colleagues identified mistrust of government agencies and uncertainty about vaccine safety as central themes [35] for Black women. Slatton and colleagues highlighted how historical trauma and medical racism influenced vaccine refusal, while Zelaya and colleagues documented how trust in local and community-based information sources informed decisions among Black women in Kentucky [29, 37][37, 38],. Collectively, these studies emphasize how structural gendered racism shapes Black women's healthcare choices, both for themselves and their children [29]. However, to our knowledge, no study has examined Black motherhood in the broader cultural sense, particularly in relation to fictive kinship roles, such as "aunties," neighborhood matriarchs, or church mothers, in relation to COVID-19 vaccination.

Given the disproportionate burden of COVID-19 morbidity and mortality in Black communities, it is critical to understand how Black women, who serve as central caregivers and cultural leaders, navigated vaccine decision-making during the pandemic [21]. This study aims to examine how Black mothers in Michigan made vaccination choices for themselves and their families, focusing on the social, cultural, and relational factors that shaped these decisions. By situating vaccine decision-making within broader caregiving roles, community responsibilities, and systemic inequities, this study highlights how maternal leadership influenced health behaviors and responses at the community level. Guided by this framework, we ask: What social, cultural, and relational factors shaped Black womens' decisions about COVID-19 vaccination in Michigan, and how did those decisions contribute to community-level responses during the pandemic?

Literature Review

Black Vaccine Uptake and Hesitancy

Public health officials frequently highlighted vaccine hesitancy in Black communities as a barrier to controlling COVID-19 disparities [7, 14, 15, 38–40]. National surveys found that Black adults expressed greater concerns about vaccine safety than white adults, often citing the speed of development and uncertainty about long-term effects [41]. Malik and colleagues (2020) reported that Black Americans were significantly less likely to express intent to vaccinate than Whites during the early rollout period [42]. Momplaisir et al. (2021) identified mistrust of government and pharmaceutical companies, alongside structural barriers such as limited access to vaccination sites, as major determinants [38]. Similar patterns emerged in Michigan, where Cross and colleagues found persistently lower rates of vaccine confidence among Black and Latinx respondents [43]. Wagner and colleagues (2021) observed that Black Detroiters, despite high rates of COVID-19 morbidity, reported substantial skepticism toward the vaccine [44].

Although quantitative studies have identified broad patterns of vaccine hesitancy, fewer studies have examined the nuanced decision-making processes within specific groups, such as mothers. Allen and colleagues (2021) found that Black women were less likely than white women to report vaccine willingness, underscoring the role of women as central healthcare decision-makers in families [34]. Mohammed and colleagues (2023) extended these findings qualitatively, demonstrating that Black women often negotiated the competing influences of mistrust, maternal responsibility, and community well-being in their vaccine decisions [35][35],. Thornton and Reich (2022) argued that vaccine refusal among Black mothers should be understood within the broader context of gendered racism, which structures how Black women relate to the state and healthcare institutions [29]. These studies highlight the importance of examining Black maternal experiences as a distinct area of inquiry.

Black Mothering and Community Care

Patricia Hill Collins (1987) describes Black motherhood as a multifaceted practice that includes biological parenting, community mothering, and fictive kinship [28]). In many Black communities, fictive kin, such as neighborhood matriarchs, church mothers, or "aunties," are recognized as legitimate maternal figures with the authority to guide health, discipline, and social norms. This caregiving role extends beyond biological ties, making Black motherhood both a personal and communal responsibility [28]. Hill (1998) and Stack (1974) argue that this relational authority positions Black women as "othermothers," moral leaders whose guidance is deeply respected [45, 46]. In public health contexts, this maternal authority often translates into informal leadership, shaping community responses to illness, wellness practices, and health behaviors.

Inventive Mothering as Cultural Strategy

Building on this foundation, Randles (2021) introduced the concept of "Inventive Mothering" to describe the creative caregiving strategies developed by Black mothers under conditions of scarcity and systemic exclusion [47]. Unlike "Intensive Mothering," which assumes privilege and institutional support, Inventive Mothering centers race and class, prioritizing children's needs through non-institutional strategies such as home remedies, neighborhood-based networks, and protective restrictions. Historical scholarship further situates these practices within a context of justified mistrust. Roberts (1997) shows how reproductive control and medical racism have long constrained Black mothers, while Boyd (2020) and Thornton & Reich (2022) illustrate how contemporary maternal health choices continue to be shaped by structural neglect [6, 29, 48]. Within this tradition, Inventive Mothering can be understood as a cultural repertoire that allows Black mothers to protect their families even when institutional care is inadequate or mistrusted.

Self-Determination Theory and Health Motivation

Self-Determination Theory (SDT) provides a psychological framework for understanding motivation, identifying autonomy, competence, and relatedness as the three fundamental needs that shape decision-making [49]. In health contexts, SDT has been widely applied to explain behaviors such as smoking cessation, diabetes management, and physical activity, with evidence showing that autonomy-supportive environments improve adherence and outcomes [50][51],. In vaccine research, SDT has been used to explore how parental decision-making reflects tensions between autonomy and relatedness [51, 52]. Yet, SDT is often criticized for lacking cultural specificity; it assumes neutral decision-making environments rather than structurally inequitable contexts [53–56].

Integrating Frameworks

Taken together, these frameworks highlight why Black maternal leadership is a critical site of inquiry for vaccine decision-making. Black Mothering underscores the cultural authority and communal responsibility that mothers carry. Inventive Mothering illuminates the creative protective strategies mothers deploy when institutions are untrustworthy or inaccessible. SDT offers a motivational lens for categorizing decision-making processes, although its limitations necessitate cultural supplementation. Integrating these perspectives enables us to analyze vaccine decisions not only as individual health behaviors but also as culturally situated acts of maternal leadership that shape community health.

Methods

Study Design and Community Partnership

Data for Community-Centered Interventions for Improved Vaccine Uptake for COVID (CIVIC) study) were conducted between 2022 and 2023 using a community-based participatory research (CBPR) approach grounded in collaboration, shared decision-making, and mutual accountability [57]. The study partnered with the Michigan CEAL: Communities Conquering COVID (MICEAL) stakeholder table, a statewide effort funded by the National Institutes of Health (1 OT2 HL 156812. CIVIC was designed to strengthen community trust, enhance vaccine education, and reduce COVID-19 inequities across underserved populations in Michigan through equitable academic–community collaboration. Our CIVIC research team collaborated closely with the MICEAL Steering Committee (SC), comprising 16 Black and Latinx community leaders from churches, grassroots health organizations, public health departments, and mutual aid networks across Michigan.

The Partnership co-designed every significant component of this study. Specifically, the SC contributed to the development of research questions, co-created the semi-structured interview protocol, advised on culturally appropriate recruitment strategies, and approved key design decisions such as the interview incentive amount. SC members also reviewed preliminary findings and provided critical insight during theme development to ensure analytic interpretations reflected the lived realities of the communities represented.

Participant Recruitment and Eligibility

Participants were recruited through SC-led outreach, which included distributing flyers at local churches and community centers, posting on Facebook and Instagram, conducting personal outreach through MICEAL partners, and utilizing word-of-mouth snowball recruitment. To be eligible, participants had to: (1) be 18 years or older; (2) self-identify as Black/African American or Hispanic/Latinx English or Spanish speaking); and (3) be up-to-date in the vaccines (received the first two doses of the vaccine released in 2021) or not up-to-date (received only one dose of the vaccine or were unvaccinated).

Following updated CDC guidelines to reflect the new COVID-19 booster introduced in the Fall of 2022, our eligibility criteria was also updated. For interviews conducted before January 1st 2023, participants were considered

up-to-date if they had completed the primary series released in 2021 and *not up-to-date* if they had not received the primary series. Participants interviewed after January 1 st 2023 were considered *up-to-date* if they had completed the primary series from 2021 and the bivalent booster released in October 2022. They were considered *not up-to-date* if they had not received the bivalent booster.

Of 174 individuals contacted, 40 provided consent and 34 completed interviews. Among them, 24 identified as Black and 10 as Latinx, with participant ages ranging from 21 to 74 (mean = 41) (See Table 2). The current paper centers the experiences of the 20 Black-identifying women, to which led to deeper theoretical engagement with Black Motherhood and Inventive Mothering frameworks.

Interview Procedures

All interviews were conducted remotely via Zoom or by telephone, depending on the participant's preference. Each session lasted between 30 and 40 min and was audio-recorded with the participant's verbal consent. Interviews were conducted in English and were professionally transcribed. Initial participant incentives were set at $50. In early 2023, in response to guidance from the MICEAL Steering Committee (SC), incentives were increased to $75 in response to pandemic fatigue and rising inflation, ensuring equitable compensation for participants' time and insight. Adjusting incentive amounts is a common practice in qualitative and community-based research to support equitable participation and to acknowledge the evolving demands placed on communities during extended public health crises [58–60]. Our analysis reached a point of saturation after approximately 30 interviews, at which no new codes, themes, or insights emerged. Due to the rapid and adaptive nature of COVID-19 research, only limited demographic data were collected to minimize participant burden and ensure timely data collection during the pandemic See Table 1.

Table 1 Demographics

	N	%
County		
Wayne	9	45%
Washtenaw	6	30%
Kent	2	10%
Other	3	15%
Age		
18–29	3	15%
30–59	12	60%
60+	5	25%
Vaccination status*		
Up-to-date	10	50%
Not up-to-date	10	50%

* Before 1/1/23, participants were considered up-to-date if they had completed the primary series released in 2021 and not up-to-date if they had not received the primary series. After 1/1/23, participants were considered up-to-date if they had completed the primary series from 2021 and the bivalent booster released in October 2022. They were considered not up-to-date if they had not received the bivalent booster [36]

Data Analysis

Data analysis was conducted in Dedoose version 9.0.17 and further used a constructivist qualitative tradition using Braun and Clarke's reflexive thematic analysis approach (2006, 2021). This tradition assumes that meaning is co-constructed between researchers and participants and is shaped by context, an orientation consistent with community-based participatory research (CBPR) principles using Braun and Clarke's thematic analysis framework [61]. In Phase 1, three researchers independently coded three transcripts, then worked in pairs to refine a shared codebook. Weekly meetings ensured consensus and alignment with CBPR principles. Members of the SC engaged with the data and aided in interpretation throughout the process (in addition to their involvement in forming research questions, protocols, and participant recruitment). The remaining transcripts were coded inductively to identify emergent themes. In Phase 2, a deductive approach was employed, utilizing Self-Determination Theory (SDT), with a focus on the concepts of autonomy, competence, and relatedness [62]. This approach allowed us to interpret maternal vaccine decisions through both emergent themes and established theoretical frameworks. Inter-rater reliability was assessed using pooled Cohen's Kappa to evaluate coding consistency across three raters. Each pair of raters independently coded a subset of transcripts, and pairwise Cohen's Kappa values were computed for each pairing. These values were then averaged to produce a pooled coefficient ($\kappa = 0.85$), indicating strong agreement and coding reliability [63]. Regular consensus meetings were held to resolve discrepancies, refine code definitions, and ensure interpretive consistency throughout the analysis process. Analytical rigor was supported through reflexive team discussions, member checks with Steering Committee representatives, and memoing to track analytic decisions. This process included critical input from Steering Committee members such as Dr. Sarah Bailey and Dr. Erica Marsh, both Black women, as well as input from the broader study team. Reflexivity extended beyond analysis into dissemination: Dr. Bailey and Charles Williams (who is a Black man) co-presented preliminary findings at the American Public Health Association (APHA) Annual Meeting, where these interpretive nuances were further discussed with academic colleagues and a stakeholder advisory group that also included Black women leaders. Ethics approval was obtained from the University of Michigan's Health and

Behavioral Sciences Institutional Review Board (IRB Protocol #HUM00198399).

Results

Interviews with 20 Black women in Michigan revealed that vaccine decision-making was rarely an individual calculation. Rather, it was deeply relational, rooted in caregiving responsibilities, protective strategies, and the need to assert control over family and community health in the face of uncertainty and mistrust. Across accounts, mothers emphasized how their decisions, whether to vaccinate or to decline, were acts of leadership that reverberated throughout their households and extended networks. Four main themes were extracted from the data: (1) Family & Community Protection; (2) Weighing Risks & Uncertainty; (3) Autonomy & Collective Rights; and (4) Trust in Family & Distrust in Systems (See Table 2 for a Thematic Summary of Black Mothers' COVID-19 Vaccine Decision-Making). See Fig. 1 for a Thematic integration of Black mothers' COVID-19 vaccine decision-making.

Theme 1: Family & Community Protection

For many participants, decisions about COVID-19 vaccination were not individual health choices but acts of care embedded in family and community relationships. Mothers described themselves as protectors, often prioritizing the well-being of children, elders, and loved ones over their own doubts or fears. Decisions to vaccinate, or to adopt alternative protective strategies, were motivated by a deep sense of moral and cultural responsibility to safeguard others.

This ethic of protection was most clearly expressed through maternal enforcement of safety norms within families and social circles. A grandmother recounted telling her adult son, "If you're going to keep going out, then you need to get that shot. You like to be around people, so you need to protect yourself and us" (P2, up-to-date, 2022). Another mother explained that collective accountability extended beyond households: "We decided in our group, no vaccination, no trip" (P16, up-to-date, 2023).

Even those who chose not to vaccinate emphasized their continued sense of responsibility to others, often reframing care as prevention through isolation, masking, or avoiding

Table 2 Thematic Summary of Black Mothers' COVID-19 Vaccine Decision-Making

Theme	Sub-Themes	Illustrative Quotes	Community Impact
Family & Community Protection	• Sacrificial protection of children/elders • Enforcing household/community norms • Collective maternal authority	"I did not want it at first, but because I have a child with Down syndrome … I needed to be protected" (P3, up-to-date, 2022). "We decided in our group, no vaccination, no trip" (P16, up-to-date, 2023).	Maternal authority extended into families, sororities, and churches, shaping broader norms.
Weighing Risks & Uncertainty	• Awareness of comorbidities • Skepticism about vaccine speed & safety • Confusion over conflicting information	"When I looked at the fact that I was in a high risk … I felt good about my decision to try to protect myself" (P12, up-to-date, 2022). "I really don't know too well how [the vaccine] works. I just know it's scary" (P5, not up-to-date, 2022).	Risk assessment balanced between fear of COVID-19 and mistrust of institutions.
Autonomy & Collective Rights	• Bodily autonomy • Resistance to mandates • Linking vaccine to broader struggles for civil rights	"My body is my body. … If the vaccine was the right decision for you, then you have that right" (P32, not up-to-date, 2023). "It started with the vaccine, now they're telling women what to do with their bodies" (P39, not up-to-date, 2023).	Even autonomy claims were framed collectively—"not just me, but us"—connecting personal choice to structural struggles.
Trust & Distrust in Systems	• Desire to align with family decisions • Perception of unequal distribution (e.g., perceptions of the single-dose Johnson & Johnson vaccine being distributed more in Black communities)	"I just wanted to take the [vaccine] that everybody in my family had" (P38, up-to-date, 2022). "They were giving Johnson & Johnson in the Black community … that just built up a distrust" (Participant, not up-to-date, 2022).	Systemic mistrust reinforced the role of mothers as gatekeepers of protective strategies.

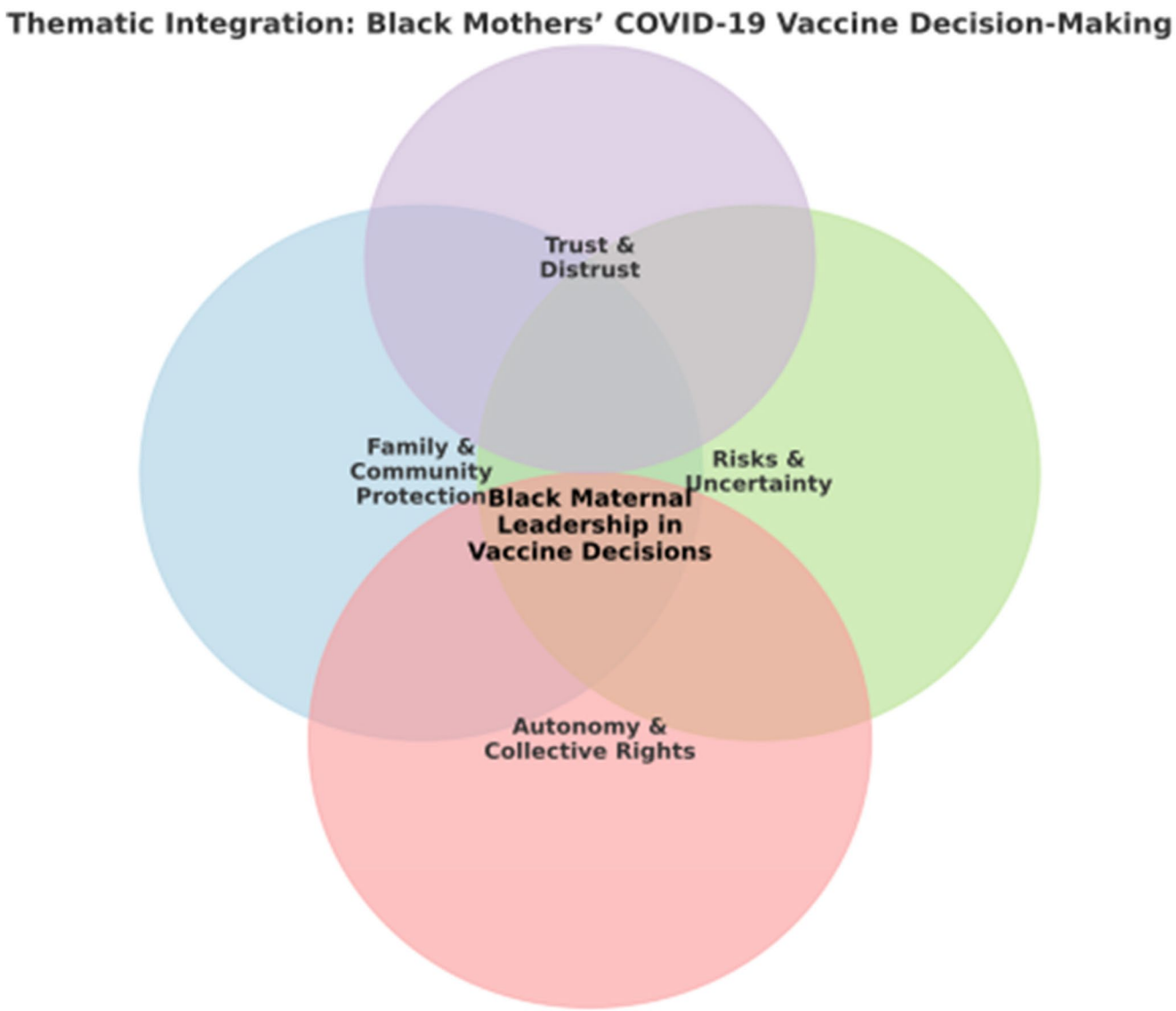

Fig. 1 Thematic Integration: Black Mothers' COVID-19 Vaccine Decision-Making. This figure illustrates the interconnection among four core themes, Family & Community Protection, Weighing Risks & Uncertainty, Autonomy & Collective Rights, and Trust & Distrust in Systems, that together center on Black Maternal Leadership in Vaccine Decisions. Each domain represents a distinct yet overlapping dimension of how Black women in Michigan navigated caregiving, moral responsibility, and systemic inequities when making vaccine decisions for their families and communities

gatherings. As one participant put it, "If you personally decide not to do it, you do have a responsibility to wear a mask, to isolate if you feel sick. That is your social responsibility" (P19, not up-to-date, 2023).

These accounts reflect what Collins (1987) describes as "othermothering", a culturally rooted form of community caregiving that extends beyond the nuclear family. Mothers in this study enacted leadership not only by protecting their own households but also by setting expectations for health behaviors within broader kinship and social networks. Through both acceptance and refusal, they modeled a relational ethic of protection that centered collective care as the foundation of community health.

Theme 2: Weighing Risks & Uncertainty

While many mothers approached vaccination as a collective act of protection, their decisions were also marked by an ongoing process of weighing personal and familial risks against widespread uncertainty. Participants described navigating an information landscape filled with fear, skepticism, and limited trust in medical and governmental sources. For some, risk was understood biologically, linked to age, comorbidities, and vulnerability. For others, it was moral and emotional, balancing fear of side effects against the possibility of leaving children without a mother.

Several mothers spoke of their decisions as pragmatic choices made under pressure. One 66-year-old woman reflected, "When I looked at the fact that I was in a high risk, and I had every risk… I felt good about my decision to try to protect myself" (P12, up-to-date, 2022). Others expressed uncertainty, citing limited or conflicting information, especially during the vaccine's early rollout. "I really don't know too well how [the vaccine] works," one participant admitted. "I just know it's scary. And I do not want it at all" (P5, not up-to-date, 2022).

For many mothers, vaccine uptake was ultimately framed as a form of sacrificial protection. One woman who initially opposed vaccination explained that she "did not want it at first," but ultimately decided to get vaccinated because her son with Down syndrome "needed to be protected. So in order for him to be protected, I needed to be protected" (P3, up-to-date, 2022). Another participant recalled the turning point that came when her children tested positive at school before vaccines were available:

"Mainly because I have children, I didn't want to leave. I didn't want something to happen to me, and then nobody's here for them. … When my kids tested positive and there was no medicine or treatment, I felt like my only defense mechanism as a mother was to get them vaccinated." (P4, up-to-date, 2022).

These narratives demonstrate that vaccine acceptance was less an expression of institutional trust than of maternal obligation. Black mothers' risk assessments were shaped by fear and uncertainty, but also by enduring commitments to care, continuity, and family survival. In this sense, their decision-making exemplifies the "inventive mothering" framework, where protective action arises not from faith in systems, but from the creative moral calculus of mothers determined to defend their families in the face of systemic neglect.

Theme 3: Autonomy & Collective Rights

Even as many mothers made decisions rooted in care for others, participants repeatedly emphasized the importance of maintaining autonomy in health decision-making. Vaccine choices were framed not merely as personal preferences, but as declarations of agency within systems perceived as coercive, inequitable, and historically untrustworthy. Mothers' narratives reflected what Self-Determination Theory describes as autonomy, the desire to act freely and on one's own terms, yet their understanding of autonomy was deeply relational rather than individualistic.

For some, asserting choice meant taking the vaccine after careful consideration, not because of public mandates or medical pressure. As one woman put it, "I still come to my own decision. And that's the way I do it. I come up with my own after I've heard" (P3, up-to-date, 2022). Others described bodily autonomy in explicitly moral or political terms. "My body is my body," one participant stated firmly. "If the vaccine was the right decision for you, then you have that right. I don't need to talk it over with anybody" (P32, not up-to-date, 2023).

Still others connected vaccine debates to broader struggles over women's rights and bodily control, reflecting awareness of structural constraints that extend beyond the pandemic. One mother explained, "It started with the vaccine, now they're telling women what to do with their bodies. … You can't start taking away people's choices, because then we are not a free society anymore" (P39, not up-to-date, 2023). For these participants, the assertion of choice was not simply resistance to vaccination, it was resistance to being overruled.

At the same time, participants' expressions of autonomy often coexisted with collective responsibility. Even those who rejected vaccination described efforts to protect others through masking or isolation, reinforcing that their choices were shaped by a sense of accountability, not disregard. One mother summarized this tension between autonomy and care: "I might not trust what they're saying, but I still care about my people. I stay home if I'm sick. That's my way of protecting them" (P19, not up-to-date, 2023).

A few mothers expressed exhaustion with ongoing vaccine debates, illustrating how agency sometimes meant disengagement. A 66-year-old grandmother explained, "When people are stuck on stupid, you're wasting your breath. … I don't think it's my job to tell them. So, when people are stuck on stupid, I let them stay there" (P12, up-to-date, 2022). Her statement reflects both fatigue and resolve, a decision to preserve peace while upholding her own convictions.

Across narratives, autonomy was not oppositional to care but intertwined with it. Mothers balanced the moral weight of protecting others with the necessity of preserving their own right to choose. Their stories reveal that autonomy, for Black mothers, is rarely exercised in isolation, it is expressed through a constant negotiation between me and us, between protecting one's family and resisting systems that threaten to control their bodies, voices, and communities.

Theme 4: Trust in Family & Distrust in Systems

Across interviews, mothers described a durable pattern: they trusted family and close networks more than formal systems, and they often filtered public health guidance through those intimate sources. This orientation both buffered uncertainty and shaped concrete choices about vaccination.

For some, family acted as a safety cue. One participant explained her decision simply: "I just wanted to take the one that everybody in my family had. … It felt safer to do

what they did" (P38, up-to-date, 2022). Here, kin alignment served as a proxy for credibility; matching relatives' decisions helped transform a high-stakes, high-uncertainty choice into a socially validated one.

At the same time, many mothers articulated active distrust toward institutions, including pharmaceutical companies, government agencies, and local implementation sites. One participant recalled the early rollout discourse around the Johnson & Johnson vaccine, described as less effective and facing safety concerns, and how she observed it clustered in Black neighborhoods:

"When it first started rolling out… the Johnson & Johnson one was the least effective and had the most problems, but that was the one they were rolling out in the Black community… The rec center had the Johnson & Johnson vaccine… That just built up a distrust like, 'Mm-mm. No.'" (P27, not up-to-date, 2022).

Another mother tied her fear to the sense of being used or experimented on: "They were looking for human trials. … People say, 'well, they're practicing on us.' … I had already made up in my mind that I was going to do it, but I just was scared… Mainly because I have children… I felt like [vaccinating them] was my defense mechanism as a mother" (P4, up-to-date, 2022). Her account captures the duality that ran through many narratives: a protective turn toward family coupled with wariness of system-level actors, even when she ultimately chose vaccination.

Together, these accounts show how kin trust and system distrust coexisted, and how both could lead to protective action. Some mothers followed family cues toward vaccination; others rejected institutional guidance yet still enacted robust protective norms (masking, isolation, household rules). In each case, decisions were legitimated through trusted relational authority, not institutional confidence: what "we" decided as a family or community carried more weight than what "they" said as a system.

Thematic Integration

As illustrated in Fig. 1, Black mothers' COVID-19 vaccine decisions were not isolated acts of individual choice but expressions of maternal leadership situated at the intersection of family, community, and structural forces. The four interrelated themes, Family and Community Protection, Weighing Risks and Uncertainty, Autonomy and Collective Rights, and Trust and Distrust, reveal how caregiving, moral responsibility, and cultural resilience coexisted within conditions of inequity and mistrust. Family served as both motivation and validator, while systemic distrust shaped the contours of what protection could look like. Even when mothers emphasized autonomy, their decisions were rarely self-centered; rather, they reflected a collective ethic of care responsive to the structural realities confronting Black families. Together, these overlapping domains form a multidimensional portrait of Black maternal agency, demonstrating how mothers navigated competing pressures to protect, resist, and lead within a racially stratified public health landscape.

Discussion

This study demonstrates that Black mothers' COVID-19 vaccine decisions in Michigan were deeply relational, rooted in caregiving roles, protective strategies, and the assertion of authority within families and communities. Rather than discrete individual choices, these decisions emerged as expressions of maternal leadership that shaped not only household health practices but also community-level norms. As illustrated in Fig. 1, four interrelated themes, Family and Community Protection, Weighing Risks and Uncertainty, Autonomy and Collective Rights, and Trust and Distrust in Systems, converge at the center on Black maternal leadership in vaccine decision-making. Each domain represents a different facet of how relational, cultural, and structural forces intertwined during the pandemic.

Consistent with previous research, protection was the primary driver of mothers' decisions [30, 31]. For many, vaccination symbolized sacrificial care, a willingness to take a personal risk to safeguard children or elders, while for others, refusal was paired with alternative protective strategies such as masking, isolation, or limiting contact. These narratives echo longstanding traditions of "othermothering" and community-based caregiving [28, 64], underscoring that maternal responsibility in Black families extends well beyond biological kinship. Mothers described themselves as gatekeepers of family safety, making decisions not just for individual health but for collective well-being.

The results also demonstrate how maternal authority was enacted through boundary-setting and enforcement. Table 2 highlights that participants not only modeled protective behavior but also required it from children, partners, peers, and sorority members. These examples of "maternal gatekeeping", refusing to allow unvaccinated family members to visit or canceling social plans due to vaccine disagreements, reflect a broader moral logic of care and accountability. This aligns with prior work showing that Black women's leadership extends into community and religious contexts, where they influence health and moral behavior [24, 31].

At the same time, vaccine hesitancy cannot be understood apart from the long shadow of historical and systemic inequity. Participants' concerns about the rapid development of vaccines, inconsistent information, and unequal distribution are not unique to Black communities, many

Americans expressed similar worries, but they take on distinct significance given Black women's lived experiences with racialized medical harm and exclusion. As one participant observed, the prevalence of the Johnson & Johnson vaccine in predominantly Black neighborhoods "just built up a distrust," reflecting how institutional practices are read through the lens of historical trauma. These findings reinforce earlier work on the enduring legacies of racialized medical mistrust [6, 29].

Autonomy was another consistent theme, though rarely articulated as pure individualism. Instead, mothers framed bodily sovereignty as part of a larger moral and political struggle over rights, particularly women's reproductive freedom. Statements such as "It started with the vaccine, now they're telling women what to do with their bodies" reflect how public health interventions were interpreted through the same structural lens as gendered and racial injustice. These connections underscore scholarship situating Black maternal authority at the intersection of health, rights, and justice [65, 66]. Even in emphasizing autonomy, participants spoke in terms of "us" rather than "me," framing self-determination as a collective, not individual, enterprise.

Finally, mothers' deep trust in family and close networks functioned as both a protective mechanism and a counterweight to institutional distrust. Some participants aligned their choices with those of trusted relatives, "I just wanted to take the one that everybody in my family had", while others interpreted vaccination campaigns through suspicion born of past exploitation. Rather than representing a deficit of trust, these dynamics illustrate adaptive strategies of protection and information-gathering rooted in cultural and historical experience.

Taken together, these findings challenge dominant public health narratives that frame vaccine uptake as a matter of personal belief or access. Instead, they position Black mothers as moral and cultural leaders whose decisions are embedded in the social fabric of family and community life. As shown in Fig. 1 and summarized in Table 2, vaccine decision-making among Black mothers was multidimensional, balancing care, risk, autonomy, and distrust within a racially stratified health landscape. Recognizing these dynamics reframes vaccine engagement from a question of compliance to one of relational care and community leadership. In this light, Black maternal authority emerges not as resistance to public health but as a vital, culturally grounded infrastructure for collective well-being.

Implications for Practice and Policy

The findings of this study underscore the urgent need for public health strategies that recognize Black mothers not only as vaccine recipients but also as cultural leaders whose decisions have a profound impact on households and communities. Their authority as caregivers, boundary-setters, and trusted advisors positions them as central figures in shaping health behaviors. Public health campaigns that neglect this reality risk missing the very people most capable of building trust and influencing protective action.

First, interventions should center Black mothers as cultural anchors. Vaccine outreach efforts that engage sororities, women's groups, and church "Mother Boards" are likely to resonate more deeply than generic campaigns because they draw upon established networks of maternal authority and trust in female voices. These existing community structures should be supported not as peripheral outreach tools but as foundational components of public health infrastructure.

Second, strategies must validate culturally grounded caregiving practices. For some mothers, vaccination was one protective strategy among many; for others, masking, isolation, or traditional remedies were expressions of care born of justified mistrust. Rather than dismissing these practices as avoidance, public health practitioners should recognize them as inventive forms of protection and build them into harm-reduction messaging. Doing so affirms agency and reinforces trust.

Third, motivational approaches must move beyond coercion to emphasize care, responsibility, and relational accountability. Consistent with Self-Determination Theory, participants responded not to mandates but to appeals rooted in the protection of children, elders, and the community. Training community health workers and outreach staff to engage through a maternal lens, asking questions that foreground family roles and caregiving responsibilities, can foster more meaningful dialogue than one-size-fits-all mandates.

Finally, recognizing Black maternal authority requires structural support. While mothers often serve as the frontline of community protection, they do so under conditions of profound stress and institutional neglect. Supporting their leadership means not only amplifying their voices but also addressing the burdens they carry. Expanding neighborhood-based clinics, offering caregiving stipends, and creating peer-support networks for community mothers would help sustain the maternal leadership that already functions as a safety net in Black communities.

Taken together, these implications highlight that health equity cannot be achieved by treating Black mothers as passive recipients of public health guidance. Instead, their authority, wisdom, and caregiving practices must be acknowledged as central to designing strategies that are culturally resonant, trustworthy, and effective.

By partnering with maternal leaders, public health systems can move closer to building models of care that genuinely reflect the lived realities and cultural logics of Black

communities. Evidence from prior community-based interventions demonstrates the value of centering moral and cultural leadership within health promotion. For example, a faith-based intervention for cocaine-addicted Black women successfully leveraged church networks and Black women's moral authority to support recovery, spiritual renewal, and community reintegration [67]. Similarly, the Journey to Native Youth Health Program, developed and piloted in Northern Plains Indigenous communities, highlights how participatory frameworks that prioritize cultural identity and community voice can improve engagement and outcomes [68]. Both examples underscore the transformative potential of community-based participatory research (CBPR) approaches that honor existing systems of care and leadership within marginalized populations.

Limitations

Several limitations should be acknowledged when interpreting the findings of this study. First, while this research was guided by a community-based participatory research (CBPR) approach, the sample was relatively small ($n=20$ Black women) and geographically limited to Michigan. Although the study offers in-depth insights into maternal decision-making within Black communities, the findings are not intended to be generalizable to all Black mothers or to populations in different regions or rural contexts.

Second, the data are based on self-reported narratives obtained through qualitative interviews, which, while rich in detail, are subject to potential biases. Participants may have presented their vaccine decisions in a way that reflects socially desirable narratives or sought to affirm community values of caregiving and protection. Moreover, the emotional and political weight of the pandemic may have shaped the framing of participants' accounts in context-specific ways.

Third, although our sample included both English and Spanish-speaking participants, this manuscript focuses solely on Black mothers. Future research should explore overlapping cultural logics across Latinx, Afro-Latinx, and other caregiving communities.

Finally, the study is subject to temporal limitations. Interviews were conducted between 2022 and 2023, a period marked by shifting CDC guidelines, declining public trust, and pandemic fatigue. Participants' perspectives were shaped by the unique circumstances of that time, including the evolving scientific knowledge and changing access to vaccines. As such, the motivations and strategies shared in this study may not fully capture how maternal health leadership and vaccine attitudes shift over time, particularly in response to future public health crises or policy changes.

Conclusion

This study highlights that Black mothers' COVID-19 vaccine decisions were not isolated acts of individual preference but relational choices rooted in caregiving, protection, and community leadership. Whether vaccinating as a form of sacrificial care or refusing vaccination while adopting alternative protective strategies, mothers framed their decisions around responsibilities to children, elders, and peers. These actions reflected a form of maternal authority that extended well beyond the household, shaping community norms and reinforcing cultural practices of care.

The findings reveal that Black maternal leadership carried particular weight in the context of mistrust, structural inequities, and politicized vaccine debates. Mothers negotiated between skepticism and responsibility, autonomy and collective accountability, drawing on traditions of Inventive and Black Mothering to guide their decisions. Importantly, even assertions of autonomy were rarely individualistic; they were bound to broader struggles over rights, bodies, and the protection of communities.

By centering these voices, this study reframes vaccine decision-making as a form of Black maternal leadership rather than as a deficit of hesitancy. Public health strategies that fail to recognize this cultural authority risk alienating the very leaders most capable of guiding households and communities toward protective action. Conversely, interventions that engage and support Black mothers as trusted messengers, caregivers, and cultural anchors can build stronger, more culturally resonant pathways toward health equity.

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Declarations

Ethics Approval The study was approved by the Institutional Review Board on the University of Michigan's Health and Behavioral Science (IRB-HSBS) Institutional Review board approval HUM00198399 on September 2, 2021. All research procedures performed in CIVIC studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Consent for Publication The authors affirm that human research participants provided informed consent for publication of their data.

Consent To Participate After a comprehensive study discussion, verbal Informed consent was obtained from all study participants. Documentation of consent was waived for this study.

Conflicts of interest/Competing Interests Dr. Erica Marsh reported receiving personal fees from Myovant Sciences and Pfizer outside the submitted work. Rev. Charles Williams is a co-investigator on the study; he is also a University of Michigan student and the co-founder and officer on the board of directors for the Institute on National Social Inequities and Gaps in Health and Health Treatment (DBA "INSIGHT"). INSIGHT is a collaborator on this study. No other disclosures are reported. There are no other conflicts to report.

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